owenmurphy_murphy1
1430-2019-31
Program 3 "Convert Fahrenheit to Celsius"

110

Program Output:

Pass Documentation Check: prog3.cpp

```
Enter a degree in Fahrenheit:
68.80 68.80 Fahrenheit is 20.44 Celsius
*** Start Extra Credit ***
68.80 Fahrenheit is 293.59 Kelvin
68.80 Fahrenheit is 528.47 Rankine
68.80 Fahrenheit is 119.33 Delisle
*** End Extra Credit ***

Enter a degree in Fahrenheit:
-10.00 -10.00 Fahrenheit is -23.33 Celsius
*** Start Extra Credit ***
-10.00 Fahrenheit is 249.82 Kelvin
-10.00 Fahrenheit is 449.67 Rankine
-10.00 Fahrenheit is 185.00 Delisle
*** End Extra Credit ***

Enter a degree in Fahrenheit:
100.00 100.00 Fahrenheit is 37.78 Celsius
*** Start Extra Credit ***
100.00 Fahrenheit is 310.93 Kelvin
100.00 Fahrenheit is 559.67 Rankine
100.00 Fahrenheit is 93.33 Delisle
*** End Extra Credit ***

Enter a degree in Fahrenheit:
0.00 0.00 Fahrenheit is -17.78 Celsius
*** Start Extra Credit ***
0.00 Fahrenheit is 255.37 Kelvin
0.00 Fahrenheit is 459.67 Rankine
0.00 Fahrenheit is 176.67 Delisle
*** End Extra Credit ***
```

PASS - Convert Fahrenheit to Celsius
PASS - Extra Credit

Solution:

```
Enter a degree in Fahrenheit: 68.80
68.80 Fahrenheit is 20.44 Celsius

*** Start Extra Credit ***

68.80 Fahrenheit is 293.59 Kelvin
68.80 Fahrenheit is 528.47 Rankine
68.80 Fahrenheit is 119.33 Delisle

*** End Extra Credit ***

Enter a degree in Fahrenheit: -10.00
-10.00 Fahrenheit is -23.33 Celsius

*** Start Extra Credit ***

-10.00 Fahrenheit is 249.82 Kelvin
-10.00 Fahrenheit is 449.67 Rankine
-10.00 Fahrenheit is 185.00 Delisle

*** End Extra Credit ***

Enter a degree in Fahrenheit: 100.00
100.00 Fahrenheit is 37.78 Celsius

*** Start Extra Credit ***

100.00 Fahrenheit is 310.93 Kelvin
100.00 Fahrenheit is 559.67 Rankine
100.00 Fahrenheit is 93.33 Delisle

*** End Extra Credit ***

Enter a degree in Fahrenheit: 0.00
0.00 Fahrenheit is -17.78 Celsius

*** Start Extra Credit ***

0.00 Fahrenheit is 255.37 Kelvin
0.00 Fahrenheit is 459.67 Rankine
0.00 Fahrenheit is 176.67 Delisle

*** End Extra Credit ***
```

PASS - Convert Fahrenheit to Celsius
PASS - Extra Credit

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File: Prg3.cpp
Last Load: 09/09/2019 23:33:19

```
/*
 Author: Owen Murphy
 Assignment Title: Convert Fahrenheit to Celcius
 Assignment Description: Write a program to calculate from fahrenheit Cel
 Due Date: 09/09/2019
 Date Created: 09/04/2019
 Date Last Modified: 09/09/2019
*/
#include <iostream>
#include <cmath>
#include <iomanip>
#include <cstdlib>

/*
 Data Abstraction
 double fahren, celsius;
*/

using namespace std;
int main() {

    cout << setprecision(2) << fixed;

    double fahren, celsius;

    cout << "Enter a degree in Fahrenheit: " << endl;

    cin >> fahren;
    cout << fahren;
    cout << " " << fahren;

    // Formula and Calculate
    celsius = (fahren - 32.0) / (1.8);

    // Output calculation
    cout << " Fahrenheit is " << celsius << " Celsius" << endl;



    cout << "*** Start Extra Credit ***" << endl;

    // <Extra Credit Output>

    // Declare var
    double kelvin, rankine, delisle;

    // convert to degrees Kelvin, Rankine, and Delisle
    // Display output  (for each Kelvin, Rankine, and Delisle

    kelvin = (fahren - 32) * (static_cast<double>(5) / 9) + 273.15;
    rankine = fahren + 459.67;
    delisle = (212 - fahren) * (static_cast<double>(5) / 6);

    cout << fahren << " Fahrenheit is " << kelvin << " Kelvin" << endl;
    cout << fahren << " Fahrenheit is " << rankine << " Rankine" << endl;
    cout << fahren << " Fahrenheit is " << delisle << " Delisle" << endl;

    cout << "*** End Extra Credit ***" << endl;



    return 0;
}
```

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